

Q1. What is Artificial Intelligence (AI)?

Answer:

Artificial Intelligence (AI) is a technology that helps computers do things that normally need human thinking. AI can:

Learn from data

Answer questions

Solve problems

Make decisions

Example:

ChatGPT answering questions

Google Maps showing directions

YouTube suggesting videos

Simple Definition:

AI is making a computer think and work like a human in some tasks.

Q2. Difference Between AI, Machine Learning, Generative AI, and Agentic AI

Answer:

Artificial Intelligence (AI)

AI is a technology that helps machines think and make decisions.

Example: Siri, Alexa

Machine Learning (ML)

Machine Learning is a part of AI. It helps computers learn from data and improve over time.

Example: Netflix movie recommendations

Generative AI

Generative AI creates new things like text, images, music, and videos.

Example: ChatGPT creating answers

Agentic AI

Agentic AI can plan and complete tasks by itself.

Example: An AI assistant that schedules meetings automatically

Simple Difference:

AI = Makes machines smart

ML = Helps machines learn

Generative AI = Creates new content

Agentic AI = Does tasks on its own

Q3. Explain Any Three Open-Source AI Models Available on Hugging Face

Answer:

1. Llama 3 Made by

Meta.

Uses:

Answers questions

Writes content

Helps with coding

2. Mistral 7B Made

by Mistral AI.

Uses:

Chatbots

Text generation

Quick responses

3. Falcon

Made by Technology Innovation Institute.

Uses:

Answering questions

Writing content

Learning and research

Simple Definition:

These are free AI models that can understand and generate text.

Q4. What is the Difference Between Traditional Programming and AI-Assisted Development?

Answer:

Traditional Programming AI-Assisted
Development

Human writes all code.

AI helps write code.

Takes more time.

Takes less time.

No AI support.

AI support is available.

More manual work.

Less manual work.

Example:

If you want a calculator program:

Traditional Programming: You write every line of code yourself.

AI-Assisted Development: You ask AI to create the code, and it helps you.

Simple Definition:

Traditional programming means writing everything yourself. AI-assisted development means AI helps you write the code.

Q5. Explain AI Hallucinations with One Example

Answer:

An AI hallucination happens when AI gives a wrong answer but says it as if it is correct.

Example:

A student asks:

"Who invented the smartphone?"

The AI may give a wrong person's name even though that person did not invent it.